

Computer Vision

Term Project

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Expected schedules

- ▶ **W13: 5/11 lecture:** (Ch10) Rect. and depth + Prof. Chuang's talk + video (term project introduction)
 - ▶ TAs will announce how to find teammates and form groups on E3.
- ▶ **W14: 5/18 Quiz**
 - ▶ Four or more essay questions. (You can take an **A4-size double-sided note** with you)
 - ▶ Finalize the term-project groups (5pm, 5/18).
- ▶ **W15: 5/25 lecture:** (Ch12) Selected topics about learning-based methods
 - ▶ Submit your proposal video (before 23:59, 5/30)
- ▶ **W16: no class**
 - ▶ Browse a number of proposal videos (before 23:59, 6/5)
- ▶ **W17: 6/8 Early-bird final demo**
- ▶ **W18: 6/15 Regular final demo** (PDF report before 6/22)

Term project schedule

- ▶ **Before May 18 (5pm):** Submit your team member list , the (temporary) project topic. (website will be announced later)
 - ▶ 1-person teams: OK, and you are encouraged to find teammates on the forum on E3. (proposal: 12~18min)
 - ▶ 2- or 3-person teams: recommended. (proposal: 15~22min)
 - ▶ 4-person teams: OK, and you are encouraged to do something more extensive or deeper. (proposal: 20~26min)
- ▶ **Before May 30:**
 - ▶ Upload video presentation about your proposal and research survey
 - ▶ List your *project title, member info*.
 - ▶ Talk about your **goal/problem**, *survey of related work*, and your **expected method/framework**.
 - ▶ Allocate more time to the survey, and focus on one or two important related methods.
- ▶ **Before June 5:**
 - ▶ Students have to browse a number of videos and provide comments or ask questions.

Term project schedule (cont.)

- ▶ **June 8:** online early final demo (1.25pt bonus)
- ▶ **June 15 :** online regular final demo
 - ▶ Presentation time per group: (temporarily, 10 minutes per team)
 - ▶ Briefly introduce your *goal/problem, method, and difficulty/uniqueness*
 - ▶ Emphasize on *uniqueness*, *results/demo*, and/or *comparison*.
- ▶ **Before June 22:**
 - ▶ Upload Report (no more than 6 pages) and codes (with comments)
 - ▶ Describe the goal/problem, method, difficulty/uniqueness, results and/or comparison.
 - ▶ List libraries and open sources used in your project.
 - ▶ List the contribution (e.g. work items) of each member?
 - ▶ *If there is a confidential problem, please discuss with Teacher in advance.*

Topics (references)

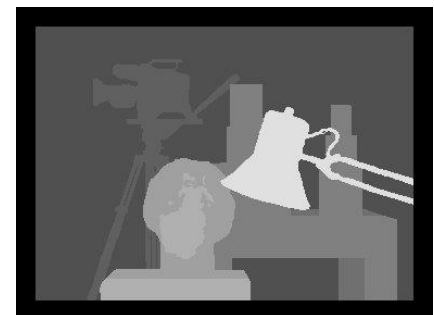
- ▶ Renowned computer vision journal and conferences:
 - ▶ Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR).
 - ▶ Proc. Intl. Conf. Computer Vision (ICCV).
 - ▶ European Conf. Computer Vision (ECCV).
- ▶ IEEE Trans. Pattern Analysis and Machine Intelligence (PAMI).
- ▶ Intl. J. Computer Vision (IJCV).

Topics (references)

- ▶ Full papers of related renowned conferences:
(Usually more than 7 pages)
 - ▶ Proc. Neural Information Processing Systems (NeurIPS)
 - ▶ Proc. Intl. Conf. Machine Learning (ICML)
 - ▶ Proc. ACM SIGGRAPH, SIGGRAPH Asia.
 - ▶ Proc. ACM Multimedia
 - ▶
- ▶ Related renowned ACM or IEEE journals,
e.g. TIP, TMM, TOG, TVCG, TCSVT, etc.

Topics (Extension of course slides)

- ▶ *Depth map from stereo images.*
- ▶ Data
 - ▶ Acquired by yourselves (calibration and rectification)
 - ▶ Middlebury <http://vision.middlebury.edu/stereo/data/>
 - ▶ Other indoor scene datasets
 - ▶ Prof. Chuang's calibration methods:
<https://github.com/jessie0915/Geometry-based-Camera-Calibration-Using-Closed-form-Solution-of-Principal-Line>
- ▶ *Methods: (you have to implement parts of the method by yourselves)*
 - ▶ Block matching
 - ▶ Dynamic programming (DP: class slides)
 - ▶ Optimization methods
 - ▶ V. Kolmogorov et al., Computing Visual Correspondence with Occlusions via Graph Cuts, Proc. ICCV'01.
 - ▶ J. Sun et al, Stereo Matching Using Belief Propagation, IEEE T. PAMI, 2003.
 - ▶ Incorporating SIFT, DNN features....

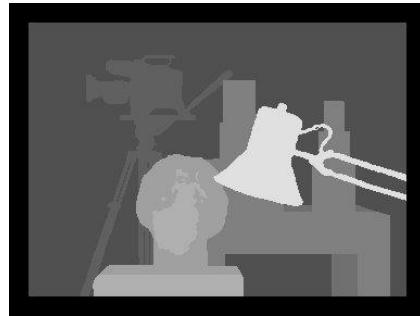


Topics (Extension of course slides)

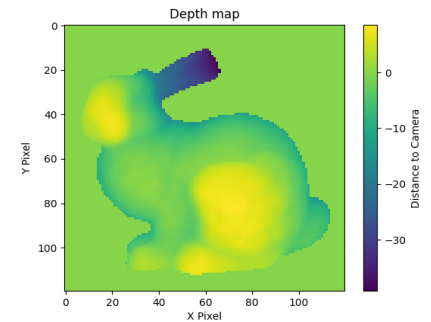
- ▶ *Depth map from stereo images + refinement with photometric stereo*



Structure from motion



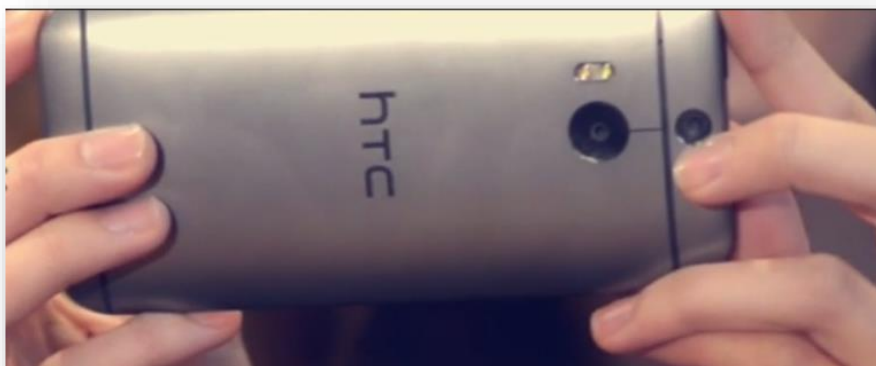
Photometric stereo



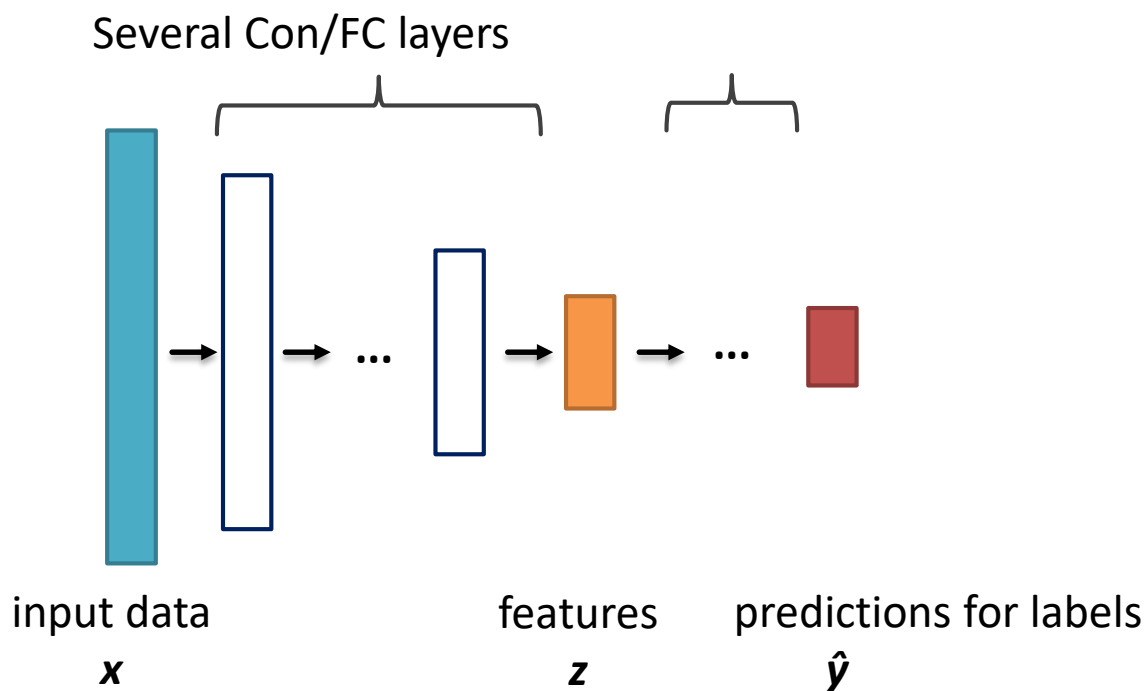
- ▶ How to take their advantages and refine the surface details ?
 - ▶ Blending the depth?
 - ▶ Conditional Filtering?
 - ▶ Optimization with certain constraints?

Topics (Extension of course slides)

- Applications with depth maps: special effects
(e.g. UFocus of HTC, Google camera, Apple portrait mode)



Topics (Incorporating modern features)



E.g. Simonyan and Zisserman, Very Deep Convolutional Networks for Large-scale Image Recognition, Proc. ICLR'15.

VGG19
input
conv3-64
conv3-64
maxpool
conv3-128
conv3-128
maxpool
conv3-256
conv3-256
conv3-256
conv3-256
maxpool
conv3-512
conv3-512
conv3-512
conv3-512
maxpool
conv3-512
conv3-512
conv3-512
conv3-512
maxpool
FC-4096
FC-4096
FC-1000
soft-max

Topics (Incorporating modern features (cont.))

- DNN features at higher layers contain some semantic information.



reference



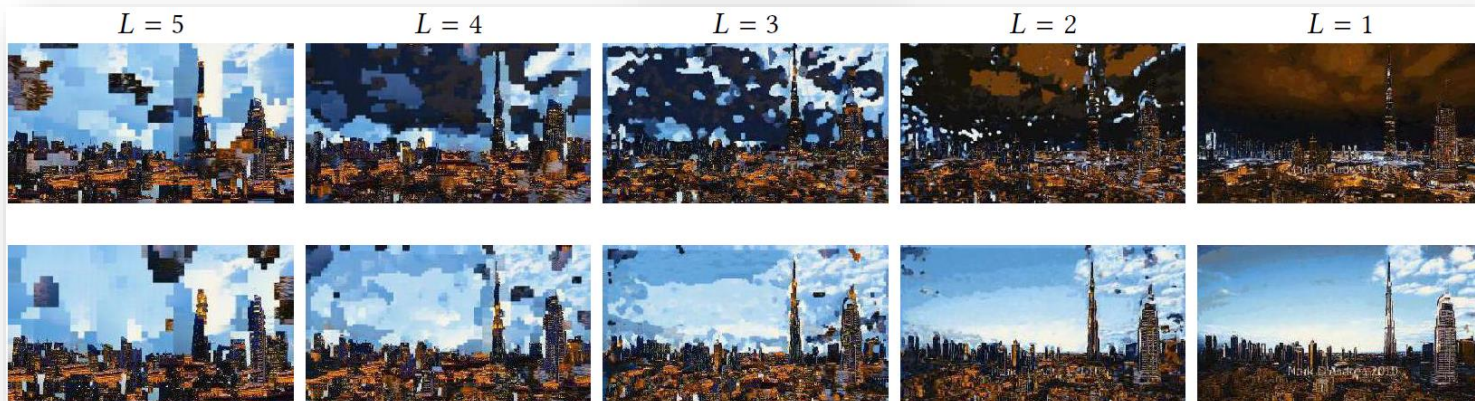
source 1



source 2

Nearest neighbor field (NNF)

$ReluL_1, L = 5, \dots, 1$ in VGG19



Topics (Incorporating modern features (cont.))

- ▶ Incorporating DNN features for object detection or tracking.
 - ▶ Comparison with SIFT or other classic features.

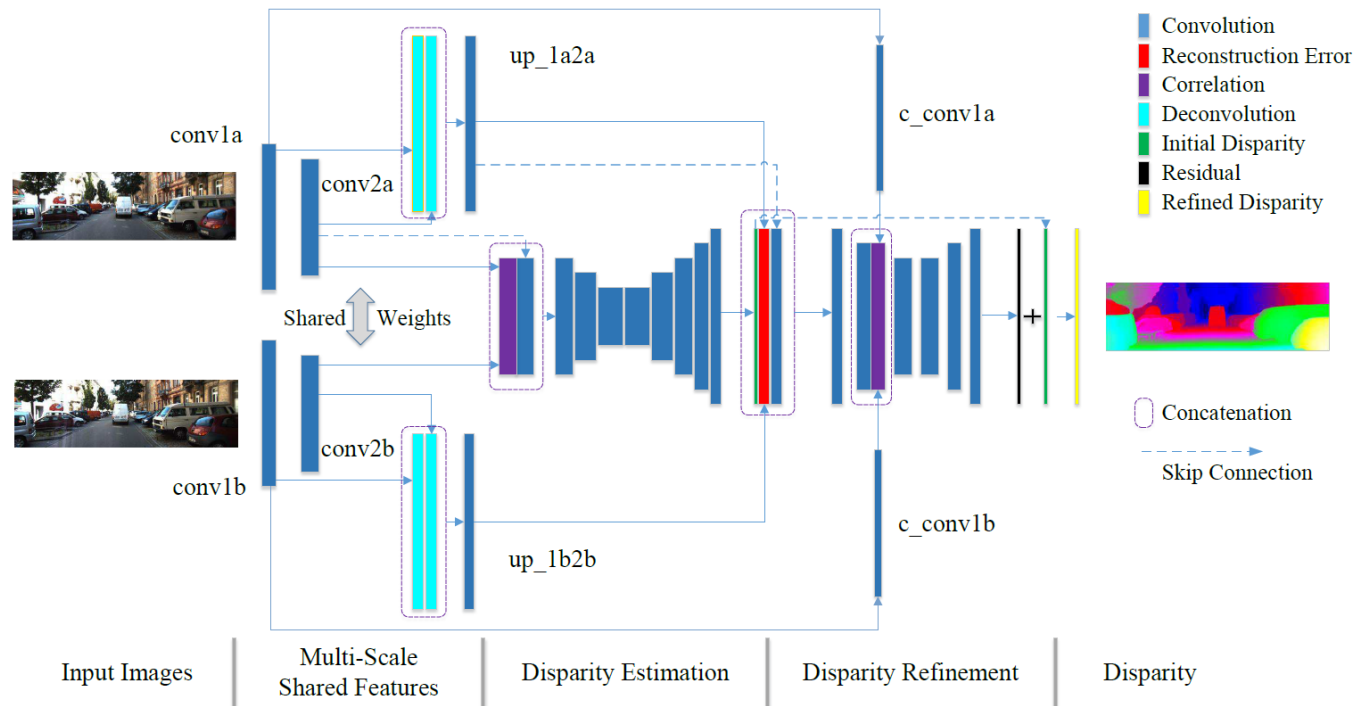


E.g.

- D. Lowe, “Distinctive Image Features from Scale-invariant Keypoints,” Intl. J. Computer Vision (IJCV), 60(2):91-110, 2004.
- Huiyu Zhou, et al., “Object tracking using SIFT features and mean shift”, Computer Vision and Image Understanding (CVIU) 2009.

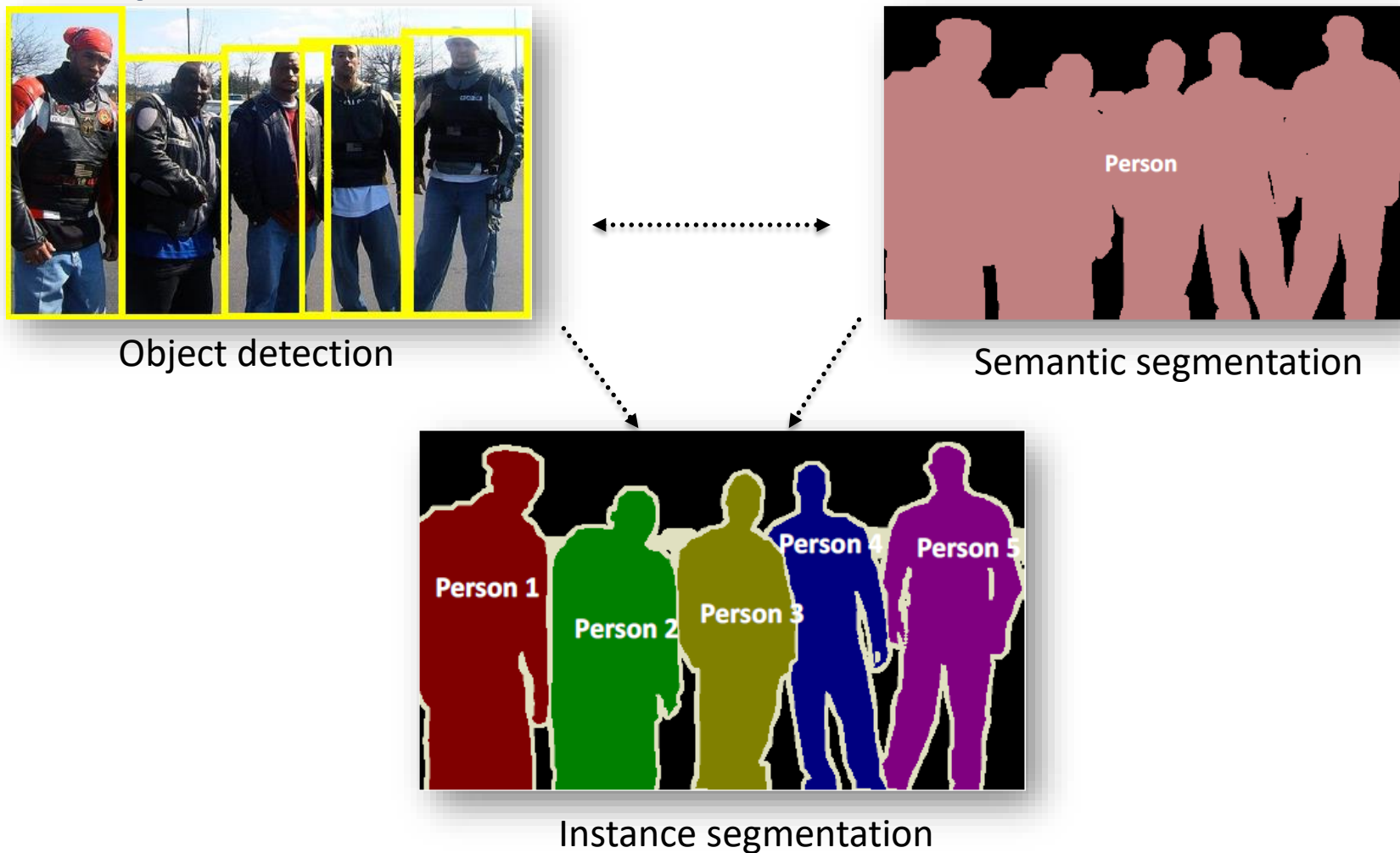
Topics (Modern approaches for classic problems)

► E.g. Disparity estimation



Liang et al, Learning for Disparity Estimation through Feature Constancy, CVPR'18.

Topics (Modern approaches)



E.g. He et al., Mask RCNN, ICCV 2017.

Wang et al., SOLOv2: Dynamic and Fast Instance Segmentation, NeurIPS 2020.

Topics (Your expertise)

- ▶ You are encouraged to work on CV-related topics that you or your lab specializes in.
- ▶ Your project can also be developed based on some work from your lab.
- ▶ However, please explicitly describe your contribution in this CV term project.

About usage of public codes/libraries

- ▶ You can use public codes or libraries, but you have to clearly **point out your sources** during your presentation and in your report.
- ▶ It is **now allowed** to hand over a project in which **only one or two pretrained models** are **directly applied**.
- ▶ You have to do at least one of the followings:
 - ▶ Enhance/modify certain parts of the codes
 - ▶ Integrate one or multiple public codes into your system **for a new goal**.
 - ▶ Compare and discuss the pros and cons of multiple methods